

Example 3: A Narrative Passage (Elinor)

FactReasoner — Ideation

June 17, 2026

1 A narrative passage (Elinor)

Source paragraph

Plain text.

Elinor is surprised no letter arrives with details about the marriage, and she wonders how Edward felt being close to Barton. Elinor asks her mother to write to Colonel Brandon for news, realizing she hoped something would prevent Edward’s marriage. Elinor feels incredibly hurt by Edward’s marriage. Colonel Brandon arrives at the house, but it is revealed to be Edward instead. Elinor is surprised that Edward and Lucy were married so soon, and she mulls over the event of Edward’s marriage, supposing that the Ferrars must be settled at Delaford, envisioning Edward and Lucy’s life there. Edward enters the room looking ill with anxiety, and Elinor tells herself to be calm upon Edward’s arrival. Elinor thinks someone in London might have informed her about Edward and Lucy’s marriage. Marianne and Margaret retreat, leaving Elinor and Mrs. Dashwood with Edward, while Marianne and Mrs. Dashwood react with shock and uncertainty about Edward’s arrival. Mrs. Dashwood breaks the ice by shaking Edward’s hand and wishing him happiness. However, no word has arrived from Colonel Brandon, and Mrs. Dashwood asks about Mrs. Ferrars’ health, to which Elinor inquires if Mrs. Ferrars is at Longstaple. Edward is shocked to discover his mother is not nearby, and Elinor pointedly asks about Mrs. Edward Ferrars. Edward is confused and asks if Elinor means Mrs. Robert Ferrars instead, leading to everyone being shocked by the revelation that Robert married Lucy Steele. Elinor flees the room, unsure of how to react, and bursts into tears of joy. Edward sits in silence, unsure of what to do, then simply leaves the room. Everyone is left perplexed by the situation.

Annotated. Each highlighted clause is one atomic unit F^n ; the rotating colours just separate adjacent units, and the pivotal revelation **F27** is orange.

Elinor is surprised no letter arrives with details about the marriage^{F1}, and she wonders how Edward felt being close to Barton^{F2}. Elinor asks her mother to write to Colonel Brandon for news^{F3}, realizing she hoped something would prevent Edward’s marriage^{F4}. Elinor feels incredibly hurt by Edward’s marriage^{F5}. Colonel Brandon arrives at the house^{F6}, but it is revealed to be Edward instead^{F7}. Elinor is surprised that Edward and Lucy were married so soon^{F8}, and she mulls over the event of Edward’s marriage^{F9}, supposing that the Ferrars must be settled at Delaford^{F10}, envisioning Edward and Lucy’s life there^{F11}. Edward enters the room looking ill with anxiety^{F12},

and Elinor tells herself to be calm upon Edward's arrival^{F13}. Elinor thinks someone in London might have informed her about Edward and Lucy's marriage^{F14}. Marianne and Margaret retreat^{F15}, leaving Elinor and Mrs. Dashwood with Edward^{F16}, while Marianne and Mrs. Dashwood react with shock and uncertainty about Edward's arrival^{F17}. Mrs. Dashwood breaks the ice by shaking Edward's hand^{F18} and wishing him happiness^{F19}. However, no word has arrived from Colonel Brandon^{F20}, and Mrs. Dashwood asks about Mrs. Ferrars' health^{F21}, to which Elinor inquires if Mrs. Ferrars is at Longstaple^{F22}. Edward is shocked to discover his mother is not nearby^{F23}, and Elinor pointedly asks about Mrs. Edward Ferrars^{F24}. Edward is confused^{F25} and asks if Elinor means Mrs. Robert Ferrars instead^{F26}, leading to everyone being shocked^{F28} by the revelation that Robert married Lucy Steele^{F27}. Elinor flees the room, unsure of how to react^{F29}, and bursts into tears of joy^{F30}. Edward sits in silence, unsure of what to do^{F31}, then simply leaves the room^{F32}. Everyone is left perplexed by the situation^{F33}.

Atomic units (self-contained, in order of appearance)

Pronouns are resolved to their referents so each unit stands alone.

- F1.** Elinor is surprised that no letter arrives with details about the marriage.
- F2.** Elinor wonders how Edward felt being close to Barton.
- F3.** Elinor asks her mother to write to Colonel Brandon for news.
- F4.** Elinor realizes she had hoped something would prevent Edward's marriage.
- F5.** Elinor feels incredibly hurt by Edward's marriage.
- F6.** A visitor arrives at the house and is initially taken to be Colonel Brandon.
- F7.** The visitor is revealed to be Edward instead of Colonel Brandon.
- F8.** Elinor is surprised that Edward and Lucy were married so soon.
- F9.** Elinor mulls over the event of Edward's marriage.
- F10.** Elinor supposes that the Ferrars must be settled at Delaford.
- F11.** Elinor envisions Edward and Lucy's life at Delaford.
- F12.** Edward enters the room looking ill with anxiety.
- F13.** Elinor tells herself to be calm upon Edward's arrival.
- F14.** Elinor thinks someone in London might have informed her about Edward and Lucy's marriage.
- F15.** Marianne and Margaret retreat from the room.
- F16.** Marianne and Margaret leave Elinor and Mrs. Dashwood alone with Edward.
- F17.** Marianne and Mrs. Dashwood react with shock and uncertainty about Edward's arrival.
- F18.** Mrs. Dashwood breaks the ice by shaking Edward's hand.

- F19.** Mrs. Dashwood wishes Edward happiness.
- F20.** No word has arrived from Colonel Brandon.
- F21.** Mrs. Dashwood asks about Mrs. Ferrars' health.
- F22.** Elinor inquires whether Mrs. Ferrars is at Longstaple.
- F23.** Edward is shocked to discover Edward's mother is not nearby.
- F24.** Elinor pointedly asks about Mrs. Edward Ferrars.
- F25.** Edward is confused by Elinor's question about Mrs. Edward Ferrars.
- F26.** Edward asks whether Elinor means Mrs. Robert Ferrars instead.
- F27.** It is revealed that Robert married Lucy Steele.
- F28.** Everyone is shocked by the revelation that Robert married Lucy Steele.
- F29.** Elinor flees the room, unsure of how to react.
- F30.** Elinor bursts into tears of joy.
- F31.** Edward sits in silence, unsure of what to do.
- F32.** Edward then leaves the room.
- F33.** Everyone is left perplexed by the situation.

Relations

Logical prerequisites ($A \rightarrow B$). In a narrative the prerequisite relation is largely *enabling causation* / *temporal grounding*: an earlier event makes a later one possible or meaningful.

- $F6 \rightarrow F7$ — a visitor must arrive before being re-identified as Edward.
- $F7 \rightarrow F12$ — Edward being the arrival is presupposed by Edward entering the room.
- $F9 \rightarrow F10$ — mulling over the marriage grounds the supposition about Delaford.
- $F10 \rightarrow F11$ — supposing the Ferrars settle at Delaford grounds envisioning their life there.
- $F15 \rightarrow F16$ — Marianne and Margaret must retreat for Elinor and Mrs. Dashwood to be left with Edward.
- $F22 \rightarrow F23$ — Elinor's question about Longstaple prompts Edward's discovery his mother is not nearby.
- $F24 \rightarrow F25$ — the pointed question about "Mrs. Edward Ferrars" is what confuses Edward.
- $F25 \rightarrow F26$ — Edward's confusion grounds his counter-question about Mrs. Robert Ferrars.
- $F26 \rightarrow F27$ — Edward's counter-question surfaces the revelation that Robert married Lucy.
- $F27 \rightarrow F28$ — the revelation is what shocks everyone.

- $F27 \rightarrow F30$ — the revelation (Edward is *not* married) is why Elinor’s tears are tears of *joy*.

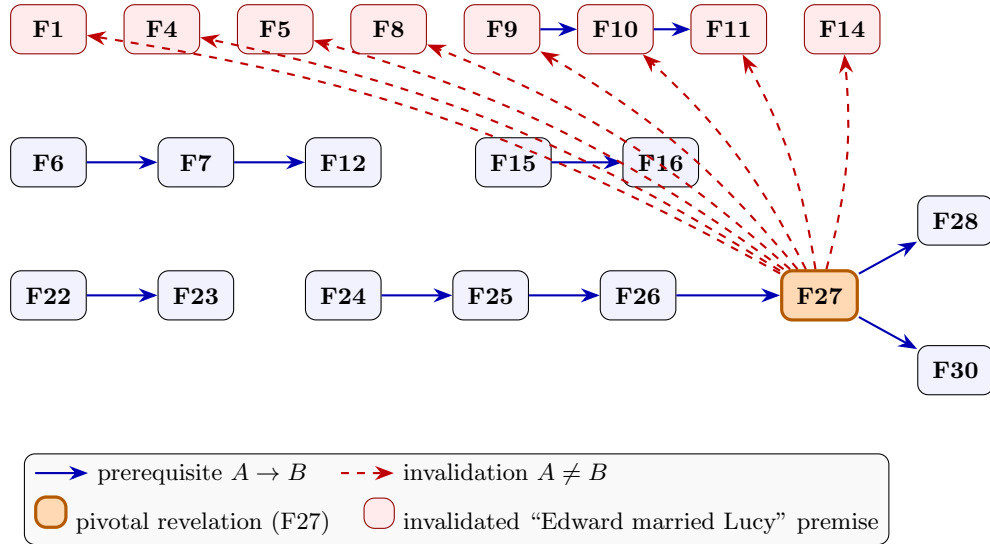
Position invalidations ($A \neq B$). The pivotal unit is **F27**: the revelation that it was *Robert* (not Edward) who married Lucy Steele. Appearing late in the passage, it retroactively invalidates the false premise – woven through the entire opening – that *Edward* married Lucy.

- $F27 \neq F1$ — the awaited “marriage” news concerned a marriage of Edward that never happened.
- $F27 \neq F4$ — there was no marriage of Edward for Elinor to have hoped to prevent.
- $F27 \neq F5$ — Elinor’s hurt rested on a marriage of Edward that did not occur.
- $F27 \neq F8$ — “Edward and Lucy were married so soon” is false; Edward did not marry Lucy.
- $F27 \neq F9$ — there was no “event of Edward’s marriage” to mull over.
- $F27 \neq F10$ — the Ferrars settling at Delaford as a married couple (Edward + Lucy) is invalidated.
- $F27 \neq F11$ — the envisioned life of Edward and Lucy at Delaford never existed.
- $F27 \neq F14$ — the supposed London report of *Edward and Lucy’s* marriage was mistaken.

Note. The invalidations all share a single source (F27) and a single target premise (“Edward married Lucy”). The events of arrival, greeting, and dialogue (F6, F7, F12–F26) are *not* invalidated – they actually happened; only Elinor’s (and the narrator’s) interpretation of *who* was married is overturned.

Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$). Dashed red arrows are position invalidations ($A \neq B$), all emanating from the pivotal revelation **F27**. The eight invalidated “Edward-married-Lucy” premises are tinted red; F27 is highlighted.



(For legibility the graph shows the principal chains and the full invalidation fan; purely sequential narration edges such as $F1 \rightarrow F2$ or $F31 \rightarrow F32$ are omitted. The mapping of these units onto the ground-truth scene below is given in Table 1, and the ground-truth dependency graph in Figure 1.)

Comparison against the ground-truth passage

The paragraph analysed above is a paraphrase of the following *ground-truth* account of the same scene. We re-decompose the ground truth, align it to the units F1–F33, and check that the relation structure (the pivot at the “Robert married Lucy” revelation, and the invalidation fan it triggers) is preserved.

Plain text.

Elinor finds herself mulling over the event of Edward’s marriage — she realizes that she’d hoped that something would come up to prevent it, but now that it’s happened, she feels incredibly hurt. She’s surprised that he and Lucy were married so soon, before he could possibly have been ordained. She wonders how he must have felt, being so close to Barton. She supposes that the Ferrars must be settled at Delaford already, and envisions their life there. Elinor had thought that someone in London might have seen fit to tell her of Edward and Lucy’s marriage, and she’s surprised when no letter arrives with details. She even asks her mother to write to Colonel Brandon, eager for any news. However, no word has arrived. As soon as she asks this, though, Colonel Brandon himself arrives — but wait, it’s not the Colonel. In fact, it’s Edward! Elinor tells herself to be calm. Marianne and Mrs. Dashwood both freak out as well, unsure of what they should do. They silently await his arrival. Edward enters, looking ill with anxiety. Mrs. Dashwood breaks the ice by shaking his hand and wishing him happiness. Marianne and Margaret both retreat, leaving Elinor and Mrs. Dashwood to deal with the visitor. Mrs. Dashwood asks if Mrs. Ferrars is well, and Elinor, following up, asks if she’s at Longstaple. Edward is shocked — his mother isn’t anywhere near here! Elinor pointedly asks about Mrs. Edward Ferrars. All eyes turn to Edward. Edward stammers, confused — does Elinor mean Mrs. Robert Ferrars instead? Everyone is shocked: what’s the meaning of this? This is the deal — it turns out that Robert, Edward’s younger brother, just married Lucy Steele. Whoa. Our minds are blown. Elinor isn’t sure how to react, so she flees the room and bursts into tears of joy. Edward also doesn’t know what to do, so he sits in silence for a moment, then simply leaves. Everyone is totally perplexed.

Annotated. Each highlighted clause is one ground-truth atomic unit G^n ; the pivotal revelation **G28** is orange. Casual interjections (“Whoa”, etc.) are not units.

Elinor finds herself mulling over the event of Edward’s marriage G^1 — she realizes that she’d hoped that something would come up to prevent it G^2 , but now that it’s happened, she feels incredibly hurt G^3 . She’s surprised that he and Lucy were married so soon G^4 , before he could possibly have been ordained G^5 . She wonders how he must have felt, being so close to Barton G^6 . She supposes that the Ferrars must be settled at Delaford already G^7 , and envisions their life there G^8 . Elinor had thought that someone in London might have seen fit to tell her of Edward and Lucy’s marriage G^9 , and she’s surprised when no letter arrives with details G^{10} . She even asks her mother to write to Colonel Brandon, eager for any news G^{11} . However, no word has arrived G^{12} . As soon as she asks this, though, Colonel Brandon himself arrives G^{13} — but wait, it’s not the Colonel. In fact, it’s Edward! G^{14} Elinor tells herself to be calm G^{15} . Marianne and Mrs. Dashwood both freak out as well, unsure of what they should do G^{16} . They silently await his arrival G^{17} . Edward enters, looking ill with anxiety G^{18} . Mrs. Dashwood breaks the ice by shaking his hand and wishing him happiness G^{19} . Marianne and

Margaret both retreat, leaving Elinor and Mrs. Dashwood to deal with the visitor ^{G20}. Mrs. Dashwood asks if Mrs. Ferrars is well ^{G21}, and Elinor, following up, asks if she's at Longstaple ^{G22}. Edward is shocked — his mother isn't anywhere near here! ^{G23} Elinor pointedly asks about Mrs. Edward Ferrars ^{G24}. All eyes turn to Edward ^{G25}. Edward stammers, confused — does Elinor mean Mrs. Robert Ferrars instead? ^{G26} Everyone is shocked: what's the meaning of this? ^{G27} This is the deal — it turns out that Robert, Edward's younger brother, just married Lucy Steele ^{G28}. Whoa. Our minds are blown. Elinor isn't sure how to react, so she flees the room and bursts into tears of joy ^{G29}. Edward also doesn't know what to do, so he sits in silence for a moment, then simply leaves ^{G30}. Everyone is totally perplexed ^{G31}.

Ground-truth atomic units (self-contained, in order of appearance).

- G1. Elinor mulls over the event of Edward's marriage.
- G2. Elinor realizes she had hoped something would come up to prevent Edward's marriage.
- G3. Elinor feels incredibly hurt now that Edward's marriage has happened.
- G4. Elinor is surprised that Edward and Lucy were married so soon.
- G5. Edward and Lucy were married before Edward could possibly have been ordained.
- G6. Elinor wonders how Edward must have felt being so close to Barton.
- G7. Elinor supposes that the Ferrars must be settled at Delaford already.
- G8. Elinor envisions the Ferrars' life at Delaford.
- G9. Elinor had thought someone in London might tell her of Edward and Lucy's marriage.
- G10. Elinor is surprised when no letter arrives with details of the marriage.
- G11. Elinor asks her mother to write to Colonel Brandon for news.
- G12. No word has arrived from Colonel Brandon.
- G13. As soon as Elinor asks, a visitor arrives and is taken to be Colonel Brandon.
- G14. The visitor is in fact Edward, not Colonel Brandon.
- G15. Elinor tells herself to be calm.
- G16. Marianne and Mrs. Dashwood are alarmed and unsure what to do.
- G17. Marianne and Mrs. Dashwood silently await Edward's arrival.
- G18. Edward enters the room looking ill with anxiety.
- G19. Mrs. Dashwood breaks the ice by shaking Edward's hand and wishing him happiness.
- G20. Marianne and Margaret retreat, leaving Elinor and Mrs. Dashwood with the visitor.
- G21. Mrs. Dashwood asks whether Mrs. Ferrars is well.

- G22.** Elinor asks whether Mrs. Ferrars is at Longstaple.
- G23.** Edward is shocked because Edward’s mother is not anywhere near Barton.
- G24.** Elinor pointedly asks about Mrs. Edward Ferrars.
- G25.** All eyes turn to Edward.
- G26.** Edward, confused, asks whether Elinor means Mrs. Robert Ferrars instead.
- G27.** Everyone is shocked and asks the meaning of this.
- G28.** Robert, Edward’s younger brother, just married Lucy Steele.
- G29.** Elinor, unsure how to react, flees the room and bursts into tears of joy.
- G30.** Edward, unsure what to do, sits in silence and then leaves.
- G31.** Everyone is totally perplexed.

Alignment to F1–F33. Table 1 maps each predicted unit to its ground-truth counterpart. The two decompositions agree on the scene’s content; the ground truth additionally makes explicit two facts the paraphrase dropped — that the marriage preceded Edward’s ordination (**G5**) and that Robert is *Edward’s younger brother* (sharpening **G28** vs. predicted F27).

Ground-truth dependency graph. Figure 1 draws the relations on the ground-truth units, using the same conventions as the predicted graph above. The structure is preserved: **G28** (Robert married Lucy) is the pivot, and it invalidates exactly the units that presuppose “Edward married Lucy” (G1–G4, G7–G10) — the same premise set as the predicted F27 fan. The extra ground-truth fact **G5** (“married before ordination”) is likewise invalidated, since the marriage it describes is Edward’s.

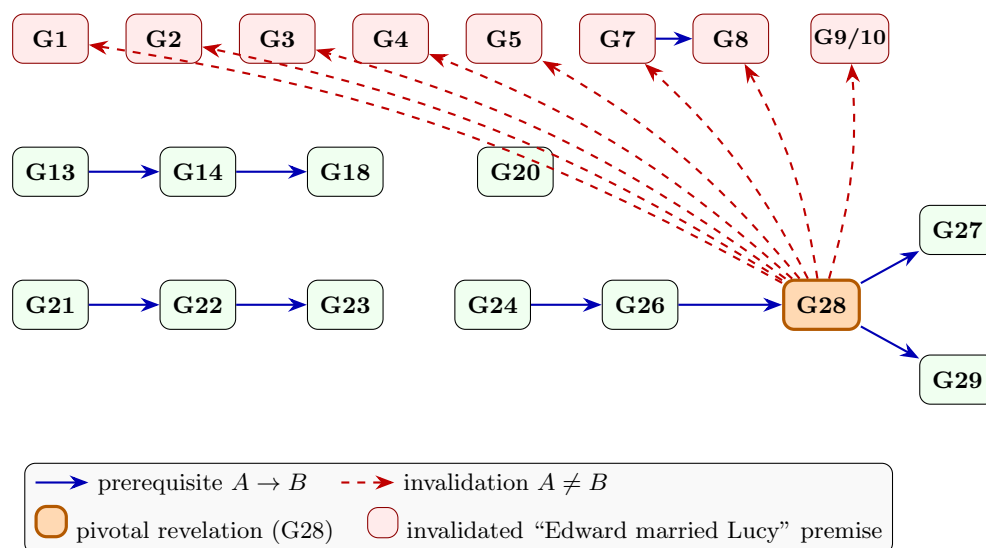


Figure 1: Ground-truth dependency graph (units G1–G31). The pivot **G28** mirrors the predicted pivot F27 (Table 1); its invalidation fan covers the same premise set, plus the ground-truth-only unit G5 (married before ordination).

Predicted	Ground truth	Note
F1, F14	G10, G9	awaited / missing news of the marriage
F4	G2	hoped to prevent it
F5	G3	feels hurt
F8	G4	surprised married so soon
F9	G1	mulls over the marriage
F10, F11	G7, G8	Delaford supposition / envisioned life
F2	G6	wonders about Barton
F3	G11	asks mother to write to Brandon
F20	G12	no word from Brandon
F6, F7	G13, G14	“Brandon” arrives → is Edward
F12	G18	Edward enters, ill with anxiety
F13	G15	Elinor tells herself to be calm
F17	G16, G17	others alarmed; silently await
F18, F19	G19	Mrs. Dashwood greets / wishes happiness
F15, F16	G20	Marianne & Margaret retreat
F21	G21	asks if Mrs. Ferrars is well
F22	G22	asks if at Longstaple
F23	G23	Edward shocked: mother not near
F24	G24	pointed “Mrs. Edward Ferrars”
F25, F26	G26 (+G25)	confused; “Mrs. Robert Ferrars?”
F27	G28	<i>pivot</i> : Robert (Edward’s brother) married Lucy
F28	G27	everyone shocked
F29, F30	G29	Elinor flees; tears of joy
F31, F32	G30	Edward silent, then leaves
F33	G31	everyone perplexed
–	G5	<i>ground-truth only</i> : married before ordination

Table 1: Alignment of the predicted units F1–F33 to the ground-truth units G1–G31. Bold marks the shared pivot.

Temporal-order discrepancies

The ground-truth units **G1–G31 are listed in chronological order**: G_i happens before G_j whenever $i < j$. The altered passage, however, narrates the *same* events in a different order. Using the alignment of Table 1, we assign each predicted unit its *chronological rank* (the index of the ground-truth event it reports) and ask whether reading F1, F2, . . . , F33 in order visits those ranks in increasing order. Wherever a later predicted unit reports an *earlier* event than one before it, the narration has jumped backwards in time — a temporal discrepancy.

Chronological rank of each predicted unit.

F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
rank (G)	10	6	11	2	3	13	14	4	1	7	8	18	15	9	20	20	16
F	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
rank (G)	19	19	12	21	22	23	24	25	26	28	27	29	29	30	30	31	

In a perfectly ordered paraphrase this rank sequence would be non-decreasing. Instead it lurches: it opens at rank 10, drops to rank 2 (F4), reaches back to rank **1** at F9 (the chronological *start* of the scene, narrated ninth), and only settles into order from F21 onward. Two discrepancies survive into the otherwise-ordered tail: F20 (rank 12, the missing word from Brandon, displaced very late) and the climactic local swap F27/F28 (rank 28 before rank 27).

What is out of place.

- **Scrambled opening cluster (F1–F14).** Chronologically the scene runs G1 (mulls)→G2 (hoped to prevent)→G3 (hurt)→ G4 (so soon)→G6 (Barton)→G7/8 (Delaford)→ G9/10 (London / no letter)→G11 (asks mother)→G12 (no word). The paraphrase instead *opens* with the missing-letter surprise (F1=G10) and the Barton musing (F2=G6), asks the mother (F3=G11) *before* establishing the hurt (F4/F5=G2/G3), and reaches the true starting beat “Elinor mulls over the marriage” (F9=G1) only ninth.
- **Displaced “no word from Brandon” (F20=G12).** In the ground truth this immediately follows the request to write (G11); the paraphrase moves it past the arrival and greeting, to just before Mrs. Dashwood’s question about Mrs. Ferrars.
- **Climactic swap (F27↔F28, i.e. G28 before G27).** The paraphrase states the revelation (Robert married Lucy) and then reports the general shock; the ground truth has the shocked “what’s the meaning of this?” (G27) precede the explanatory reveal (G28).

Discrepancy plot. Figure 2 plots each predicted unit at (appearance order, chronological rank). A faithful retelling would lie on the rising dotted diagonal; every point *below* that diagonal is an event told later than its true time, and every **red back-arc** connects consecutive units where the story jumps backwards in time. The dense tangle of back-arcs over F1–F14 is the scrambled opening; the lone late arcs are F20 and the F27/F28 swap.

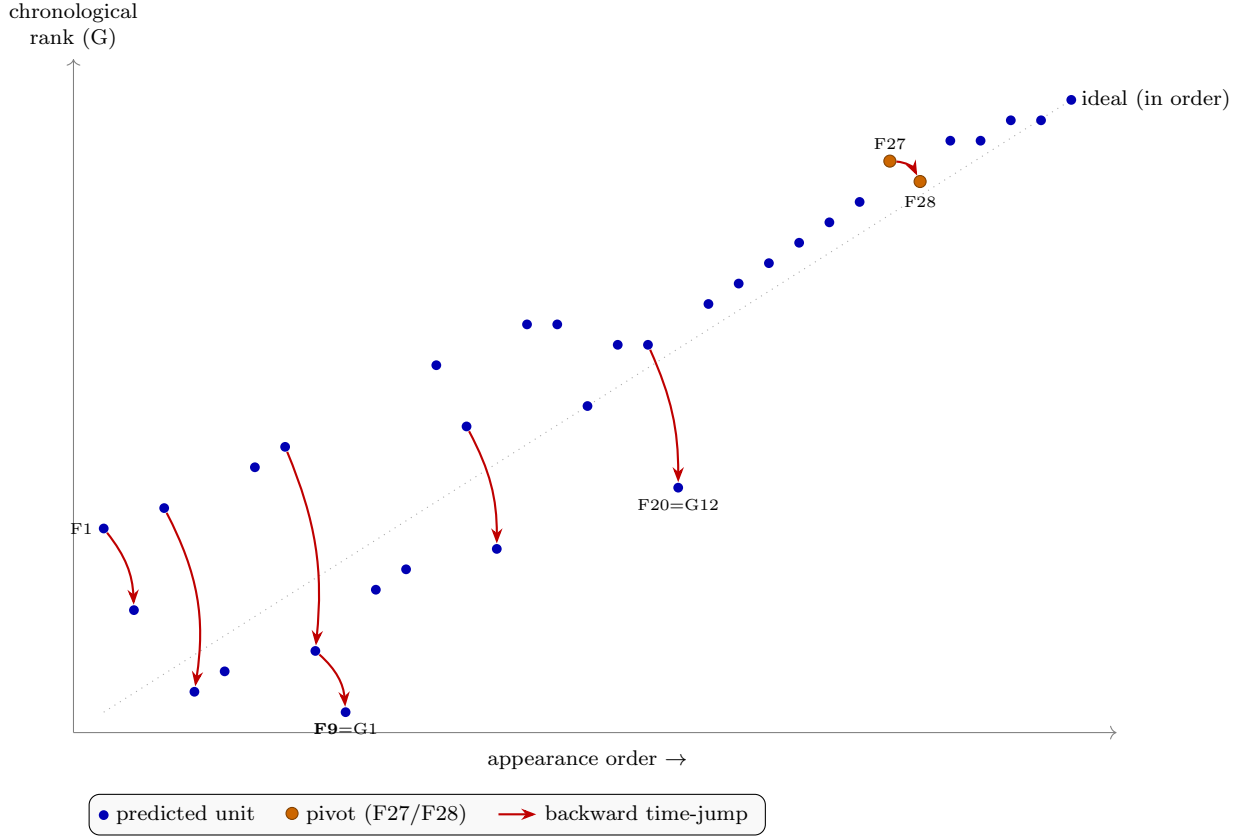


Figure 2: Temporal-order discrepancy plot. Each predicted unit F_i is placed at its appearance order (x) and the chronological rank of the ground-truth event it reports (y). A faithful retelling tracks the dotted diagonal; points below it are told late, and red arcs mark consecutive backward jumps in time. The opening cluster F_1 – F_{14} is badly scrambled (it reaches the true first event, $F_9=G_1$, only ninth), while F_{21} – F_{33} is ordered apart from the displaced F_{20} and the climactic F_{27}/F_{28} swap.